

Pure Mathematics with Mechanics 3
0765

**SOUTH WEST REGIONAL MOCK EXAMINATION
GENERAL EDUCATION EXAMINATION**

THE TEACHERS' RESOURCE UNIT (TRU)

IN COLLABORATION WITH

**THE REGIONAL INSPECTORATE OF PEDAGOGY AND
THE SOUTH WEST ASSOCIATION OF MATHEMATICS TEACHERS (SWAMT)**

FRIDAY 27/03/2026: AFTERNOON

ADVANCED LEVEL

Subject Title	Pure Mathematics with Mechanics
Subject Code No.	0765
Paper No.	3

Mobile phones are NOT allowed in the examination room.

Duration: Three Hours

INSTRUCTIONS TO CANDIDATES

Answer ALL questions.

For your guidance, the approximate mark allocation for parts of each question is indicated in brackets.

You are reminded of the necessity for good English and orderly presentation in your answers.

Mathematical formulae and tables published by the GCE Board and noiseless, non-programmable calculators are allowed.

In calculations, you are advised to show all the steps in your working, giving your answer at each stage.

1. Two identical particles P and Q each of mass 2 kg move in the x, y -coordinate plane such that at time t , their velocities are $\mathbf{v}_1$ and $\mathbf{v}_2$ respectively, where
- $$\mathbf{v}_1 = [(2t)\mathbf{i} - 3t^2\mathbf{j}] \text{ ms}^{-1} \quad \text{and} \quad \mathbf{v}_2 = [t^3\mathbf{i} + (2t - 3)\mathbf{j}] \text{ ms}^{-1}$$

Find,

- the velocity of P relative to Q when $t = 2$, (2 marks)
- the non-zero values of t when the direction of motion of P is perpendicular to Q , (2 marks)
- the force acting on P when $t = 2$, (4 marks)

Initially, both particles are at the origin, find

- the distance between the particles when $t = 2$. (5 marks)

2. (i) A uniform rod AB of length $2l$ and mass m is smoothly hinged at A on a vertical wall and kept in equilibrium in a horizontal position by a light inextensible string BC , where C is vertically above A . The string makes an angle θ with the rod, where $\tan \theta = \frac{1}{2}$, show that
- the tension in the string is of magnitude $\frac{\sqrt{5}}{2}mg$. (4 marks)
 - the magnitude of the reaction at the hinge is same as the tension in the string. (4 marks)
- (ii) A particle of mass 300 g is in circular motion round a circular path of radius 80 cm . Given that the particle completes one revolution in 1.2 s , find the centripetal force acting on the particle. (5 marks)

3. Two particles A and B of respective masses $\frac{11}{2}m$ and $\frac{9}{2}m$ are connected by a light inextensible string passing over a smooth fixed pulley. The system is released from rest with both particles vertical and the string taut, when A is at a distance d from the horizontal ground.

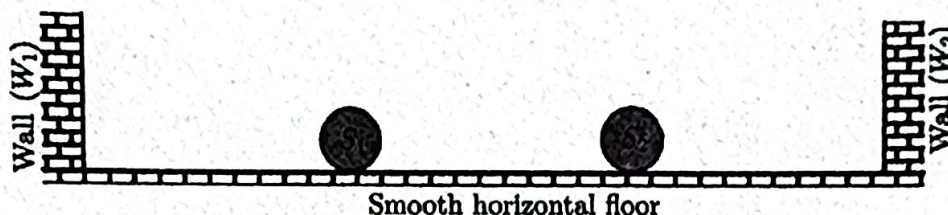
Find

- the acceleration of the system and the tension in the string. (5 marks)
- the magnitude of the reaction force at the pulley. (2 marks)

If a time t elapses before A hits the ground,

- show that $20d = gt^2$. (3 marks)
- find the speed with which A strikes the ground. (3 marks)

4. Two smooth spheres S_1 and S_2 of equal radii but of masses 1 kg and 2 kg respectively lie in a straight line on a smooth horizontal floor between two smooth vertical walls W_1 and W_2 as shown in Figure 1 below.



Sphere S_2 is projected with speed 6 ms^{-1} towards the wall W_2 so that it collides directly, rebounds and collides with sphere S_1 . Sphere S_1 subsequently moves on to collide directly with wall W_1 . Given that the coefficient of restitution in any collision is $\frac{1}{3}$, find

- the speeds of S_1 and S_2 immediately after they collide. (8 marks)
- the magnitude of the impulse exerted by S_1 on S_2 . (2 marks)

Show that the speed of sphere S_1 immediately after colliding with wall W_1 is $\frac{16}{27} \text{ ms}^{-1}$. (3 marks)

5. A particle, P is projected from a point O on horizontal ground with velocity $u(2\mathbf{i} + 3\mathbf{j}) \text{ ms}^{-1}$, where $\mathbf{i}$ and $\mathbf{j}$ are unit vectors along OX and OY respectively. P strikes the plane at the point A , where $OA = 750 \text{ m}$ from O .

- Show that $u = 25$. (3 marks)
Hence, find

- the maximum height above the level of projection reached by P . (3 marks)

- the time of flight by the particle from O to A . (3 marks)

The particle P is moving with speed 65 ms^{-1} when it strikes a bird in its path,

- Find the distance of the bird above the ground at the time it was struck by P . (4 marks)

6. (i) A lorry of mass 5 tonnes traveling in a straight line at a constant speed of 15 ms^{-1} up a road inclined at $\sin^{-1}\left(\frac{1}{10}\right)$ to the horizontal. The non-gravitational resistance to the motion of the lorry is of magnitude 2500 N, find the rate at which the engine of the lorry is working. (5 marks)

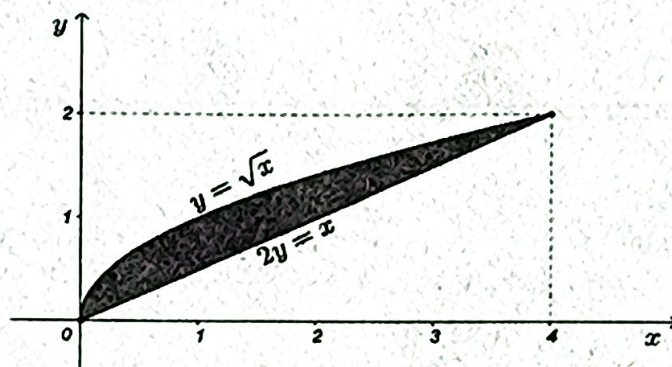
- (ii) An elastic string PQ of natural length 1.6 m lies along the line of greatest slope of a plane inclined at 60° to the horizontal. The end P is fixed while a particle of mass 2 kg is attached to the end Q so that the length of the string is now 2 m and the system is in equilibrium. Find

- the modulus of elasticity of the string, (5 marks)

- the elastic potential energy stored in the string. (3 marks)

7. (i) A system is made up of four particles of masses 2 kg, 3 kg, 2 kg and 1 kg placed in the x, y -coordinate plane at the points $(2, 3)$, $(-1, 0)$, $(1, 2)$ and $(-2, -3)$ respectively. Find the coordinates of the centre of mass of the system. (5 marks)

- (ii) The shaded region in Figure 2 below defines a uniform lamina bounded by the curves $y = \sqrt{x}$ and $2y = x$



Find the coordinates of the centroid of the lamina.

(8 marks)

8. (i) The events A and B are such that $P(B|A) = \frac{3}{8}$, $P(A|B) = \frac{4}{9}$ and $P(B|A') = \frac{15}{28}$.

Find $P(A)$.

(5 marks)

- (ii) One hundred students were sampled in the city of Kumba on a particular school day. 15 went to school by taxi, 55 on motorbikes and the rest on foot. Given that 2% of those who went by taxi, 7% of those who went by motorbikes and 5% of those who went on foot arrived school late that day. One student was randomly chosen, find the probability that

(a) the student is late.

(3 marks)

(b) the student arrived school on time.

(2 marks)

A student was chosen at random and was found to arrive school late that day.

(c) find the probability that the student went to school on foot.

(3 marks)